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To the Editors, *Journal of Systems and Software*

Re: Submission — “Protocol-Governed Computing: A Software Development Lifecycle Architecture for Deterministic Declarative Execution and Governed Transformation”

Dear Editors,

I submit the manuscript above for consideration as a regular research paper.

What the paper is about. Software behavior is rarely determined in one place. Some is stated in declarations; some is supplied by build tooling, runtime defaults, environment discovery, or fallback behavior. The paper treats this as a question of where behavioral authority resides across the lifecycle, and presents an architecture in which one semantic schema governs all three activities: transformation determines the next declarations from a named predecessor, construction determines whether candidate declarations may exist and emits a sealed snapshot, and execution realizes that snapshot without adding behavioral authority. Each transition produces evidence; a refused transition produces no usable governed result. The architecture is then reduced to a compact set of checkable conformance obligations and assessed against a named, publicly deposited reference realization.

Why JSS. The contribution is a systems-and-software architecture with an explicit conformance and evidence framing, not a language design, a formal-methods result, or a runtime optimization. It sits where JSS has long published: software architecture and evolution, declarative and model-driven construction, reproducibility and provenance, and the evaluation of realizations against stated obligations. The paper is also deliberately bounded in the way JSS reviewing rewards — it separates what the architecture requires, what the specification makes normative, what the implementation was observed to do, and what the evidence establishes, and it names the claims the evidence does not support. The realization is deposited and reproducible by a runbook, so the empirical portion is open to inspection rather than asserted.

What is novel. The lifecycle composition itself is a synthesis of prior deposited work, and the paper says so. The novel contributions are: unifying governed transformation, governed construction, and deterministic execution under a single predecessor-to-successor transition while preserving the distinct authority of each; deriving a compact conformance view — closure, reachability, inspection, refusal, evidence, externally authored profiles — from that composition; and, most importantly, a discipline of claim separation that prevents the presence of a governance mechanism from standing in for evidence that the mechanism secures the property it was built for. Checks that are red by design or advisory are reported as such rather than resolved into a binary maturity claim.

I am an independent researcher writing from more than forty years in large-systems integration as a systems architect. The problem addressed here is one I met repeatedly in practice long before it had a name: systems that work while no single artifact determines what they may do. That vantage is what the paper contributes in place of a laboratory.

The manuscript is original, is not under consideration elsewhere, and has not been published previously. A companion study by the same author, cited in §1.3 for disclosure, is under concurrent review elsewhere; it addresses a different question and reuses no text, argument, or evidence from this paper. Prior PGC preprints synthesized here are deposited and cited by DOI. There is no funding and no conflict of interest. Use of a generative AI assistant is disclosed in the manuscript.

Thank you for your consideration.

Sincerely,

Bhash Ganti